PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: TradeBridge ASEAN: Cross-Border SME Pricing & Market Entry Intelligence

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian micro, small, and medium enterprises (MSMEs) lack the sophisticated analytical power to dynamically price their products across borders. They struggle to navigate fluctuating exchange rates, opaque cross-border logistics fees, varying import duties, and localized competitor pricing. Relying on static markups often leads to eroded profit margins or uncompetitive pricing that results in market rejection.
- **Target Domain:** Cross-border e-commerce pricing intelligence and automated market entry strategy for the ASEAN region.
- **Proposed Solution Summary:** TradeBridge ASEAN removes the friction of cross-border scaling by providing an automated, AI-driven market-entry and pricing engine. Utilizing n8n workflows and the Z.AI GLM-5.1 model, the system daily scans ASEAN markets, computes landed costs, and proposes reasoned pricing adjustments 1-by-1 to the business owner via a Telegram bot for asynchronous approval.

2. Background & Business Objective

- **Importance of Solving This Issue:** Accelerating MSME expansion across ASEAN requires transitioning from manual, error-prone spreadsheet calculations to proactive market optimization. By automating the calculation of volatile variables (e.g., Thai VAT, Philippine de minimis thresholds, MYR currency fluctuations), merchants can confidently maintain baseline profitability and competitive parity without the cognitive burden of daily manual research.
- **Strategic Fit / Impact:** The solution aligns with the broader ASEAN Digital Economy Framework Agreement (DEFA) 2026 initiatives, which aim to harmonize digital trade rules and unlock up to USD 2 trillion in digital economy growth by empowering MSME

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participation.1 It also leverages an AI-native operational loop, showcasing the practical application of advanced cognitive reasoning in daily business operations.

3. Product Purpose

3.1. Main Goal of the System: To provide an intelligent, highly automated cross-border pricing intelligence layer that acts as a financial co-pilot for MSMEs.

3.2. Intended Users (Target Audience):

- Malaysian MSME Owners/Founders (F&B, retail, local manufacturers).
- E-commerce Strategy Managers.
- Logistics and Supply Chain Analysts.

4. System Functionalities

4.1. Description: The system operates as an asynchronous, event-driven orchestration pipeline built natively on n8n. It serves as a high-speed generative reasoning engine that synthesizes structured inventory data (from Google Sheets) with real-time unstructured market data (from web searches) to generate actionable financial decisions.

4.2. Key Functionalities:

- **Intelligent Market Scanning:** Daily automated retrieval of localized logistics costs, import duties, and competitor pricing.
- **Sequential HITL Approvals:** Individual price change recommendations sent via Telegram with interactive approve/reject capabilities.

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- **Strategic Market Delisting:** Autonomous capability to suggest unlisting a product in a specific country if shipping costs mathematically negate profitability

- **.Conversational Command Interface:** Telegram bot integration allowing users to trigger workflows, update inventory attributes, and list items dynamically.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI GLM-5.1 is the exclusive cognitive reasoning engine. It was selected for its 200,000-token context window , native web search tool integration , and state-of-the-art capability in long-horizon agentic workflows.

4.3.2. **Prompting Strategy:** The system employs highly constrained, zero-shot agentic prompting. The model is explicitly instructed to prioritize stability (retaining the existing price) and only propose a change if margins fall below a strict threshold or competitor parity is lost.

4.3.3. **Context & Input Handling**: To manage input sizes and reduce the USD 1.40/M token input cost , the workflow utilizes semantic batching via n8n's "Split in Batches" node. Additionally, Z.AI's Context Caching is leveraged for static system instructions and repetitive categorical web search results, which can reduce token consumption by over 80%.

4.3.4. **Fallback & Failure Behavior:** If GLM-5.1 hallucinates or breaks the requested JSON schema, the n8n structured output parser automatically routes to a failure branch. The system will apply a penalty, attempt a single retry, and if it fails again, default to a safe state (skipping the item and preserving the legacy price) while notifying the user via Telegram.

5. User Stories & Use Cases

- **User Stories:**

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- *"As an SME Owner, I want to receive 1-by-1 Telegram messages with reasoned price adjustments so that I can approve them easily without doing manual calculations."*
- *"As an E-commerce Manager, I want the system to automatically retrieve Thai VAT (7%) and Philippine de minimis (PHP 10,000) data so that my landed costs are accurately calculated."*
- *"As a Logistics Analyst, I want the AI to suggest 'unlisted' if shipping costs exceed the product's value so that we don't accidentally operate at a loss in a specific country."*

- **Use Cases (Main Interactions):**
  - **Daily Scan Flow**: n8n chronologically fetches Google Sheets inventory -> fetches external forex APIs -> triggers GLM-5.1 for web search and reasoning -> evaluates if a change is needed.
  - **Human-in-the-Loop (HITL) Approval Flow**: If a change is proposed, n8n sends a Telegram message with reasoning and inline buttons. The n8n execution thread enters a "Wait for Webhook" state. The user taps "Approve" -> the webhook resumes the workflow -> n8n writes the new price to Google Sheets.
  - **On-Demand Update Flow**: User texts /run_update in Telegram -> bot triggers the n8n webhook -> system performs a targeted scan of the inventory.

6. Features Included (Scope Definition)

- Bidirectional database synchronization via Google Sheets API (MVP data persistence layer).
- Automated daily scheduling and manual Telegram webhook triggers.

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- Cognitive web scraping utilizing GLM-5.1's native capabilities.
- Asynchronous Telegram approval loops with interactive inline keyboards.
- Automated stability notifications ("market scanned, no changes needed").

7. Features Not Included (Scope Control)

- Direct e-commerce API integration (e.g., syncing directly to live Shopee/Lazada storefronts).
- Vision processing capabilities (e.g., creating items from uploaded photos).
- Production-grade backend relational databases (e.g., PostgreSQL).
- Multilingual NLP parsing for user commands (Telegram UI remains strictly English).

8. Assumptions & Constraints

- **LLM Cost Constraint:** Evaluating 500 items across 10 ASEAN nations daily presents high inference costs (GLM-5.1 input is $1.40/M tokens). The system relies heavily on the architectural decision to implement Context Caching and group items by category (e.g., executing one web search for "Thai electronics import duties" and caching the context for 50 similar items) to keep costs manageable.
- **Technical Constraints:** Google Sheets API limits write operations strictly to 60 requests per user per minute. The n8n workflow must utilize pacing delays to prevent API quota exhaustion.
- **Performance Constraints:** The asynchronous Telegram approval relies on n8n webhook suspension. An absolute timeout constraint of 12 hours is enforced; unapproved changes will gracefully default to rejection to prevent server memory deadlocks.
- **User Input:** The model relies on the primary key (Item Name) matching exactly between Telegram commands and the Google Sheet rows.

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9. Risks & Questions Throughout Development

- **Unique Identifier Collision:** How will the system maintain database integrity if a user manually inputs duplicate "Item Names" directly in the Google Sheet spreadsheet, bypassing the Telegram bot?

- **Web Search Reliability:** E-commerce platforms employ sophisticated anti-bot countermeasures. Can the GLM-5.1 Web Search tool consistently bypass these firewalls to accurately scrape competitor prices on localized platforms?

- **Execution Latency:** Given the sequential looping, API pacing delays, and webhook wait times, will the master daily update workflow exceed maximum execution time limits imposed by standard n8n hosting environments?

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